

Chapter 2 Exploring Data with Tables and Graphs

Frequency Distributions for Organizing and Summarizing Data (2.1)

Objectives

In this lesson, we will learn how to

- organize large data sets using frequency distributions;
- identify class limits, class boundaries, class midpoints, and class width;
- construct a frequency distribution from raw data;
- create and interpret relative frequency distributions;
- create and interpret cumulative frequency distributions;
- compare two data sets using relative frequencies;
- recognize important features of a distribution, including shape and gaps.

1. Why Do We Need Frequency Distributions?

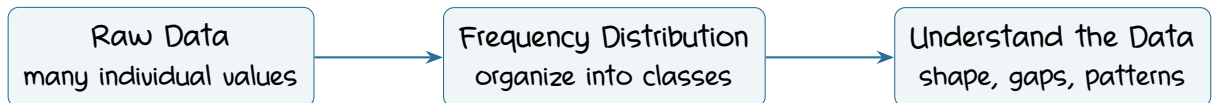
When working with a large data set, it can be difficult to see the overall pattern by looking only at the raw values.

A frequency distribution organizes the data so that important patterns become easier to recognize.

Frequency Distribution

A frequency distribution, also called a frequency table, shows how data are divided among several categories or classes.

For each class, it gives the number of observations belonging to that class. That number is called the frequency.



2. Important Vocabulary

Parts of a Frequency Distribution

Lower class limits

The smallest numbers that can belong to each class.

Upper class limits

The largest numbers that can belong to each class.

Class boundaries

Numbers used to separate classes without leaving gaps between them.

Class midpoints

Values located in the middle of the classes.

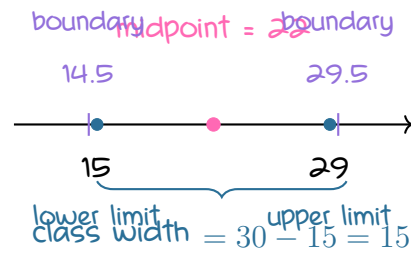
Class width

The difference between two consecutive lower class limits (or two consecutive lower class boundaries).

Understanding One Class

Consider the class

15 – 29.



Therefore,

$$\begin{aligned}\text{Lower class limit} &= 15, \\ \text{Upper class limit} &= 29, \\ \text{Lower class boundary} &= 14.5, \\ \text{Upper class boundary} &= 29.5.\end{aligned}$$

The midpoint is

$$\frac{15 + 29}{2} = 22.$$

Two Useful Formulas

$$\text{class midpoint} = \frac{\text{lower class limit} + \text{upper class limit}}{2}$$

$$\text{class width} = \text{difference between consecutive lower class limits}$$

3. Constructing a Frequency Distribution

Procedure

1. Select the number of classes, usually between 5 and 20.
2. Calculate the approximate class width:

$$\text{class width} \approx \frac{\text{maximum value} - \text{minimum value}}{\text{number of classes}}.$$

3. Round the result to a convenient number, usually rounding up.
4. Choose the first lower class limit. Use either the minimum value or a convenient value below it.
5. Add the class width repeatedly to obtain the remaining lower class limits.
6. Determine the corresponding upper class limits.
7. Place each data value into its appropriate class using tally marks, then count the tallies to obtain the frequencies.

4. Example: Commute Times in Los Angeles

Example 1:

The following are daily commute times, in minutes, for Los Angeles. Construct a frequency distribution using 7 classes.

18 25 45 75 60 40 25 8 50 10 10 30 15 25 50 20 30
 20 45 30 60 30 20 15 30 60 30 15 35 40 5 30 40 20
 10 45 30 15 25 25 5 90 30 15 60 20 60 30 25 25

Solution

Step 1: Select the number of classes.

We use

7 classes.

Step 2: Calculate the class width.

The minimum is 5 and the maximum is 90.

$$\frac{90 - 5}{7} = \frac{85}{7} \approx 12.14.$$

Round upward to a convenient value:

$$\boxed{\text{class width} = 15}.$$

Step 3: Choose the first lower class limit.

Choose

0,

because it is convenient and is below the minimum value 5.

Step 4: Find the remaining lower class limits.

Add 15 repeatedly:

0, 15, 30, 45, 60, 75, 90.

Step 5: Find the upper class limits.

Lower Limit	Upper Limit
0	14
15	29
30	44
45	59
60	74
75	89
90	104

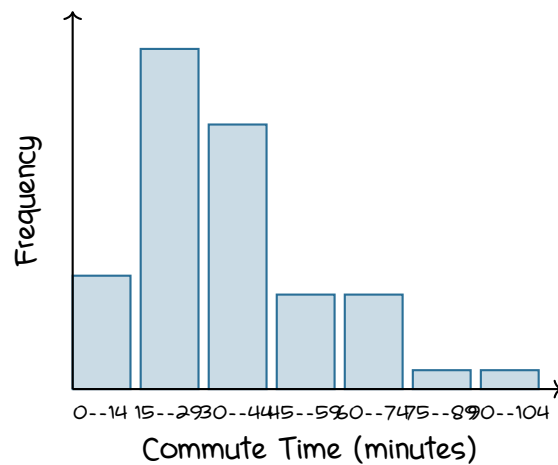
Step 6: Tally the observations and count them.

Commute Time	Tally	Frequency
0 – 14		6
15 – 29	tally observations	18
30 – 44	tally observations	14
45 – 59		5
60 – 74		5
75 – 89		1
90 – 104		1

Thus,

Daily Commute Time (minutes)	Frequency
0 – 14	6
15 – 29	18
30 – 44	14
45 – 59	5
60 – 74	5
75 – 89	1
90 – 104	1

A Visual Preview of the Distribution



Important

This graph is only a visual preview.
We will formally study and construct histograms in Section 2.2.

5. What Can We Learn from the Distribution?

What We Can See

- The class 15 – 29 has the largest frequency.
- Most commute times are less than 45 minutes.
- Very long commute times are relatively uncommon.
- The distribution is concentrated toward the lower commute times, with a smaller number of large values.

Interpretation Question

What does the frequency 18 in the class 15 – 29 mean?

Solution

It means that

18

of the 50 observed commute times are between 15 and 29 minutes, inclusive.

6. Relative Frequency Distribution

Sometimes two data sets contain different numbers of observations.

Comparing their raw frequencies may therefore be misleading.

A relative frequency distribution replaces each class frequency with a proportion or percentage.

Relative Frequency

$$\text{relative frequency} = \frac{\text{class frequency}}{\text{total frequency}}$$

For percentage frequency,

$$\text{percentage} = \frac{\text{class frequency}}{\text{total frequency}} \times 100\%$$

Example 2:

Complete the relative frequency distribution for the Los Angeles commute times.

Commute Time	Frequency	Relative Frequency
0 – 14	6	
15 – 29	18	
30 – 44	14	
45 – 59	5	
60 – 74	5	
75 – 89	1	
90 – 104	1	

Solution

There are

$$6 + 18 + 14 + 5 + 5 + 1 + 1 = 50$$

observations.

Therefore,

Commute Time	Frequency	Relative Frequency
0 – 14	6	12%
15 – 29	18	36%
30 – 44	14	28%
45 – 59	5	10%
60 – 74	5	10%
75 – 89	1	2%
90 – 104	1	2%

The percentages total

100%.

Why Use Relative Frequency?

Relative frequencies put data sets on the same scale.

This is especially useful when the data sets contain different numbers of observations.

7. Comparing Relative Frequency Distributions

Combining relative frequency distributions makes comparisons much easier.

Example 3:

The following relative frequency distributions are based on 1000 commute times in New York and Boise.

Daily Commute Time	New York	Boise
0 – 14	8.6%	30.3%
15 – 29	20.3%	45.5%
30 – 44	24.8%	17.0%
45 – 59	17.2%	3.5%
60 – 74	18.5%	2.2%
75 – 89	3.3%	0.3%
90 – 104	4.7%	0.3%
105 – 119	0.1%	0.0%
120 – 134	0.0%	0.0%
135 – 149	2.5%	0.9%

What does the table suggest?

Solution

For Boise, the first two classes contain

$$30.3\% + 45.5\% = 75.8\%.$$

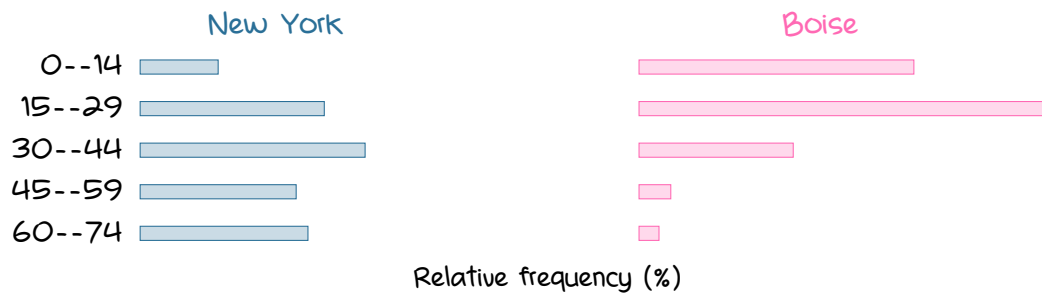
For New York, the first two classes contain only

$$8.6\% + 20.3\% = 28.9\%.$$

Therefore,

Boise commute times tend to be shorter than New York commute times.

Visual Comparison



Look Before Calculating

Which city has much more of its distribution concentrated in the lowest commute-time classes?

Solution

Boise. Its first two classes contain 75.8% of the observations, compared with only 28.9% for New York.

8. Cumulative Frequency Distribution

A cumulative frequency distribution shows running totals.

Cumulative Frequency

The cumulative frequency for a class is the frequency for that class plus the frequencies of all preceding classes.

Example 4:
Complete the cumulative frequency distribution.

Commute Time	Cumulative Frequency
Less than 15	
Less than 30	
Less than 45	
Less than 60	
Less than 75	
Less than 90	
Less than 105	

Solution

Start with the frequencies

6, 18, 14, 5, 5, 1, 1.

Now compute the running totals:

Commute Time	Cumulative Frequency
Less than 15	6
Less than 30	24
Less than 45	38
Less than 60	43
Less than 75	48
Less than 90	49
Less than 105	50

What Does a Cumulative Frequency Mean?

The cumulative frequency for “Less than 45” is 38.
What does this mean in context?

Solution

It means

38 of the 50 commuters

have commute times of less than 45 minutes.

Quick Interpretation

Cumulative frequency often answers a question of the form

“How many observations are below this value?”

9. Critical Thinking: Understanding Shape

Frequency distributions can help us recognize the general shape of a data set.
One especially important pattern is an approximately normal distribution.

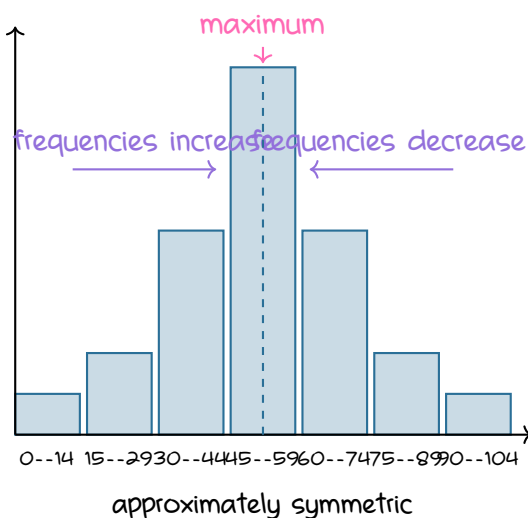
Pattern Suggesting an Approximately Normal Distribution

A frequency distribution has an approximately bell-shaped pattern when

- frequencies begin relatively low;
- frequencies increase toward one or two high frequencies;
- frequencies then decrease again;
- the two sides are approximately symmetric.

Consider the frequencies

2, 4, 10, 18, 10, 4, 2.



Key Observation

The pattern is approximately

low → high → low

and the two sides are roughly mirror images.

10. Gaps in Frequency Distributions

Sometimes one or more classes have frequency 0.

This creates a **gap** in the frequency distribution.

Gaps

A large gap can suggest that the data come from two or more different populations.

However, the converse is not necessarily true:

data from different populations do not always produce visible gaps.

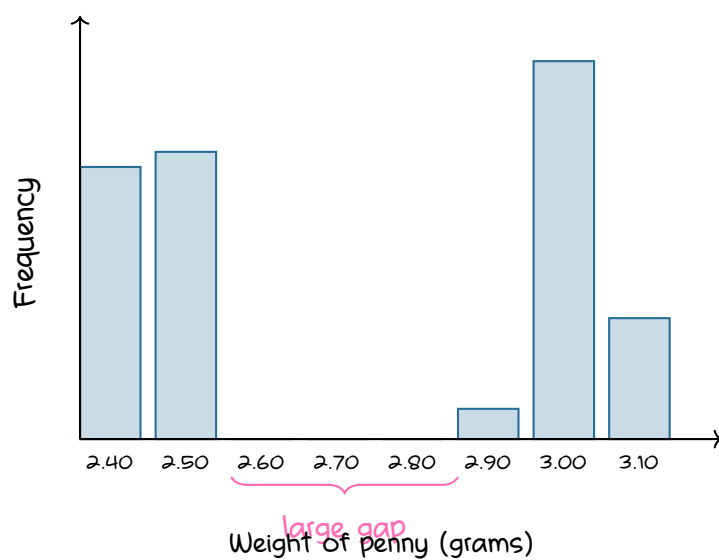
Example 5:

The following table gives the weights, in grams, of randomly selected

pennies.

Weight of Penny (grams)	Frequency
2.40 – 2.49	18
2.50 – 2.59	19
2.60 – 2.69	0
2.70 – 2.79	0
2.80 – 2.89	0
2.90 – 2.99	2
3.00 – 3.09	25
3.10 – 3.19	8

What does the large gap suggest?



Solution

The large gap suggests that the penny weights may come from two different populations.

A useful explanation is the change in penny composition.

- Pennies made before 1983 are approximately

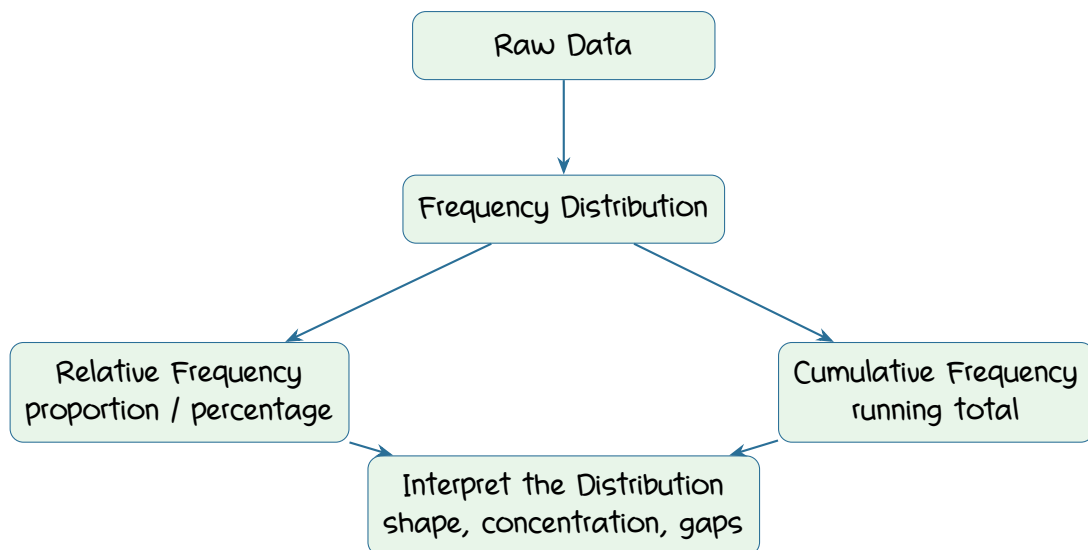
95% copper and 5% zinc.

- Pennies made after 1983 are approximately

2.5% copper and 97.5% zinc.

The different compositions produce noticeably different weights, which helps explain the large gap.

11. The Big Picture



Summary

Key Ideas

- A frequency distribution organizes data into classes and frequencies.
- Class limits are the smallest and largest values that can belong to a class.

- Class boundaries separate classes without gaps.
- A class midpoint is

$$\frac{\text{lower class limit} + \text{upper class limit}}{2}.$$

- Class width is the difference between consecutive lower class limits.
- A relative frequency distribution expresses frequencies as proportions or percentages.
- Relative frequencies are especially useful when comparing data sets.
- A cumulative frequency distribution gives running totals.
- Frequency distributions help us recognize the shape of data.
- An approximately normal distribution tends to increase toward the center and decrease afterward, with approximate symmetry.
- A large gap can suggest that observations come from different populations.